

The Grünbaum–Löwner centroid-section problem

An exact later-literature resolution

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Status. `literature_already_answered`. This is a literature-identification note, not a claim of a new proof.

The source questions

Haddad, Jiménez, and Villa, *Centroids of sections of convex bodies and Lusternik–Schnirelmann category*, arXiv:2511.22393, state two classical questions. If $K \subset \mathbb{R}^n$ is a convex body, let $\mu(K)$ be the number of hyperplanes H through its centroid $c(K)$ for which

$$c(K \cap H) = c(K),$$

and let $\mu(n) = \min_K \mu(K)$.

Grünbaum asks whether every K has at least $n + 1$ such hyperplanes. Löwner asks for the exact value of $\mu(n)$. The source records $\mu(2) = 3$ and $\mu(n) = 1$ for $n \geq 5$, and explicitly leaves dimensions $n = 3, 4$ open.

The exact later answer

Ge and Yang, *A complete solution to questions of Grünbaum and Loewner*, arXiv:2606.19865, explicitly cite arXiv:2511.22393 as reference [11]. Their Problems 1 and 2 restate the same two questions, and their Theorem 1.1 proves

$$\mu(n) = 1 \quad \text{for every } n \geq 3.$$

Combining this with the known planar value gives the full answer

$$\boxed{\mu(2) = 3, \quad \mu(n) = 1 \text{ for every } n \geq 3.}$$

Consequently, Grünbaum’s proposed $n + 1$ lower bound holds in the plane and fails in every dimension at least three.

The supporting construction is stronger than a bare counterexample. For each $n \geq 3$ it produces a C^∞ , strictly convex body with positive Gauss curvature having exactly one qualifying hyperplane.

The proof starts from a small smooth radial perturbation of the Euclidean ball. It prescribes the critical-point pattern of the half-space-volume functional using the hemispherical transform and an odd Morse function, then translates the body to its centroid while preserving that pattern by a C^2 stability estimate.

Verification and scope

The match is direct rather than inferential: the supporting paper names both questions, proves the exact missing dimensions, and cites the source paper. The source and supporting PDFs are preserved beside this note.

The source also mentions two different open directions. It asks whether in dimension $n \geq 4$ every body has *some* interior point that is the centroid of $n + 1$ sections, and it separately states the Barker–Larman section-volume uniqueness problem. Theorem 1.1 of Ge–Yang concerns sections through $c(K)$ and does not settle either of those additional questions; no claim about them is made here.

References

1. J. Haddad, C. Jiménez, and R. Villa, *Centroids of sections of convex bodies and Lusternik–Schnirelmann category*, arXiv:2511.22393 (2025).
2. X. Ge and K.-W. Yang, *A complete solution to questions of Grünbaum and Loewner*, arXiv:2606.19865 (2026).